

FRONT-END DEVELOPMENT

HACKATHON

Level-1

STUDENTNAME: ANAND P

YEAR:II

COLLEGE CODE: BRUAQ

**COLLEGE NAME: PURATCHITHALAIVI AMMA GOVERNMENT ARTS AND
SCIENCE COLLEGE - PALLADAM -641664**

DEPARTMENT : B. Sc COMPUTER

SECINCE SEMESTER:4

ROLL NO :24UCS020

REG NO :2422K2192

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TITLE:Use Case: Building a Simple Student Activity Tracker Web Application

1. Use Case and Assumptions

Students enrolled in academic courses or training programs often need a simple way to track their daily learning activities and assignments. Most existing tools are complex and require back end setup, which is difficult for beginners.

- *This project proposes a lightweight front-end web application that allows students to:*
- *View learning activities*
- *Mark activities as completed*
- *Monitor their progress*
- *Assumptions:*
- *The application is designed for beginner students.*
- *No back end or database is required.*
- *Activity data will be stored using local storage.*
- *The application will be used on both desktop and mobile devices.*

2.Problem Statement

Beginner developers face difficulty in building complete applications due to back end complexity. Students lack a simple and user-friendly tool to track their learning progress.

The goal of this project is to design and develop a front-end-only Student Activity Tracker that provides basic interactivity, clean UI design, and usability without back end integration.

3.Objective

The main objective of this project is to design a lightweight Student Activity Tracker using only front-end technologies (HTML, CSS, and JavaScript) that allows students to manage and monitor their learning activities efficiently.

4.Key Features

(List the important features that the application will provide to users.)

4.1 Home Page

Displays the application title

- *Shows a welcome message*
 - *Provides simple navigation options*
-

4.2 Activity List View

(Create a section that displays a list of activities with the following details:

- *Activity Name*
- *Short Description*
- *Current Status (Pending / Completed))*

4.3 Mark Activity as Completed

Users can click a button to mark an activity as completed.

- *The status changes from **Pending** to **Completed**.*
- *The UI updates automatically after marking completed*

4.4 Progress Summary

- *The application displays a simple progress indicator such as:*
 - *“6 out of 6 activities completed”*
 - *This helps students monitor their learning progress visually.*
-

4.5 Navigation

Navigation buttons allow users to switch between:

- *Home Section*
- *Activities Section*
- *Simple and easy-to-use interface.*

5. UI and Usability Considerations

The application ensures:

- *Clean and minimal interface*
 - *Readable typography*
 - *Consistent spacing and alignment*
 - *Clear buttons with proper labels*
 - *Responsive layout for mobile and desktop*
 - *Simple color scheme for better user experience*
-

6. Technology Used

- *The project is developed using:*
 - **HTML** – *Structure of the web application*
 - **CSS** – *Styling and responsive design*
 - **JavaScript** – *Interactivity and progress tracking*
 - **Local Storage** – *To store activity completion data*
 - *(No backend or database is used.)*
 -
-

7. Data Handling Method

- *Activity data is stored as:*
 - *A predefined JavaScript array (static data)*
 - *User updates are saved using **Local Storage***
 - *This ensures:*
 - *Data persists even after page refresh*
 - *No backend is required*
-

8. High-Level UI Flow / Wireframe

UI Flow:

1. *User opens application*
 2. *Home Page is displayed*
 3. *User clicks “Activities”*
 4. *Activity list appears*
 5. *User marks activity as completed*
 6. *Progress summary updates automatically*
-

9.Implementation Approach

- *The application is structured into three main parts:*
 - **HTML** – *Defines layout and sections*
 - **CSS** – *Styles UI and ensures responsiveness*
 - **JavaScript** –
 - *Stores activity data*
 - *Updates status*
 - *Calculates progress*
 - *Saves data in local storage*
 - *Navigation is handled using simple show/hide function*
-

Html code :

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
  <meta charset="UTF-8">
```

```
  <title>Student Activity Tracker</title>
```

```
  <link rel="stylesheet" href="style.css">
```

```
</head>
```

```
<body>
```

```
  <!-- HEADER -->
```

```
  <header class="main-header">
```

```
    <div class="container">
```

```
      <h1>Student Activity Tracker</h1>
```

```
      <p>Track Courses • Monitor Progress • Stay Organized</p>
```

</div>

</header>

<!-- NAVIGATION -->

<nav class="nav-bar">

<button onclick="showHome()">Home</button>

<button onclick="showCourses()">Courses</button>

<button onclick="addActivity()">Add Activity</button>

</nav>

<!-- HOME SECTION -->

<section id="homePage" class="page container"></section>

<!-- COURSES SECTION -->

<section id="coursePage" class="page container hidden">

<div class="card">

<h2>My Courses</h2>

<p class="description">

Mark your courses as completed and track your overall progress percentage.

</p>

<div class="progress-wrapper">

<div class="progress-info">

0% Completed

</div>

<div class="progress-bar">

<div id="progressFill"></div>

</div>

</div>

```
<div id="courseList" class="course-grid"></div>
```

```
<button class="reset-btn" onclick="resetCourses()">Reset All Courses</button>
```

```
</div>
```

```
</section>
```

```
<script src="script.js"></script>
```

```
</body>
```

```
</html>
```

CSS CODE:

```
/* GENERAL */
```

```
body {
```

```
    margin: 0;
```

```
    font-family: 'Segoe UI', sans-serif;
```

```
    background: #46505e;
```

```
    color: #333;
```

```
}
```

```
.container
```

```
{ width: 90%;
```

```
max-width: 1000px;
```

```
margin: auto;
```

```
}
```

```
/* HEADER */
```

```
.main-header {
```

background: linear-gradient(135deg, #2563eb, #1e3a8a);

color: white;

text-align: center;

padding: 40px 20px;

}

.main-header h1

{ margin: 0;

font-size: 32px;

}

.main-header p

{ margin-top:

10px; opacity:

0.9;

}

/ NAVIGATION */*

.nav-bar {

text-align: center;

background: rgb(202, 230, 82);

padding: 15px;

box-shadow: 0 2px 5px rgba(0,0,0,0.1);

}

.nav-bar button { margin:

5px; padding: 10px 20px;

border: none;

background: #2563eb;

color: rgb(245, 243, 244);

border-radius: 25px;

font-weight: 600;

cursor: pointer;

transition: 0.3s;

}

.nav-bar button:hover

{ background: #1e40af;

transform: translateY(-2px);

}

.add-activity

{ margin: 20px

0; display: flex;

gap: 10px;

}

.add-activity input { flex:

1;

padding: 10px;

border-radius: 20px;

border: 1px solid #ccc;

}

.add-activity button

{ padding: 10px

20px; border-radius:

20px;

}

/ PAGE */*

```
.page {
```

```
    padding: 40px 0;
```

```
}
```

```
.hidden
```

```
{ display:
```

```
    none;
```

```
}
```

```
/* CARD */
```

```
.card {
```

```
    background: white;
```

```
    padding: 30px;
```

```
    border-radius: 15px;
```

```
    box-shadow: 0 5px 15px rgba(0,0,0,0.08);
```

```
}
```

```
/* PROGRESS */
```

```
.progress-wrapper
```

```
{ margin: 20px 0;
```

```
}
```

```
.progress-bar
```

```
{ width: 100%;
```

```
height: 12px;
```

```
    background: #e5e7eb;
```

```
    border-radius: 10px;
```

```
    overflow: hidden;
```

```
}
```

```
#progressFill
```

```
{ height:
```

```
100%;
```

```
width: 0%;
```

```
background: linear-gradient(90deg, #22c55e, #16a34a);
```

```
transition: width 0.4s ease-in-out;
```

```
}
```

```
/* COURSE GRID */
```

```
.course-grid
```

```
{ display:
```

```
grid;
```

```
grid-template-columns: repeat(auto-fit, minmax(220px, 1fr));
```

```
gap: 15px;
```

```
margin-top: 20px;
```

```
}
```

```
.course-card
```

```
{ background:
```

```
#59b6bd; padding:
```

```
15px;
```

```
border-radius: 10px;
```

```
transition: 0.3s;
```

```
border: 1px solid #721d64;
```

```
}
```

```
.course-card:hover {
```

```
box-shadow: 0 5px 12px rgba(0,0,0,0.1);
```

```
transform: translateY(-3px);
```

```
}
```

```
.completed {
```

```
color: #16a34a;
font-weight: bold;
}

/* BUTTONS */
button {
padding: 8px 15px;
border: none;
border-radius: 20px;
cursor: pointer;
margin-top: 10px;
background: #2563eb;
color: rgb(8, 8, 8);
font-weight: 500;
transition: 0.3s;
}

button:hover
{ background: #1e40af;
}

.reset-btn {
background: #dc2626;
margin-top: 25px;
}

.reset-btn:hover
{ background: #b91c1c;
}
```

```
/* RESPONSIVE */
@media (max-width: 600px) {
  .main-header h1 {
    font-size: 24px;
  }
}
```

JAVA SCRIPT:

```
let courses = [
  { id: 1, name: "HTML Basics", completed: false },
  { id: 2, name: "CSS Fundamentals", completed: false },
  { id: 3, name: "JavaScript Essentials", completed: false }
];

if (localStorage.getItem("courses")) {
  courses = JSON.parse(localStorage.getItem("courses"));
}

// ===== HOME CONTENT =====

function loadHome()
{ document.getElementById("homePage").innerHTML = `
  <div class="card">
    <h2>Welcome</h2>
    <p>
      Add your learning activities and track completion percentage easily.
    </p>
```

```
<div class="add-activity">
```

```
<input type="text" id="homeActivityInput" placeholder="Add activity from Home">
```

```
<button onclick="addActivityFromHome()">Add Activity</button>
```

```
</div>
```

```
<p>Total Activities: <strong>${courses.length}</strong></p>
```

```
</div>
```

```
};
```

```
}
```

```
// ===== NAVIGATION =====
```

```
function showHome() {
```

```
    document.getElementById("homePage").classList.remove("hidden");
```

```
    document.getElementById("coursePage").classList.add("hidden");
```

```
    loadHome();
```

```
}
```

```
function showCourses() {
```

```
    document.getElementById("homePage").classList.add("hidden");
```

```
    document.getElementById("coursePage").classList.remove("hidden");
```

```
    displayCourses();
```

```
}
```

```
// ===== DISPLAY COURSES =====
```

```
function display Courses() {
```

```
    const list = document .get Element By Id("course List");
```

```
    list .inner HTML = "";
```

```
    courses .for Each(course =>
```

```
        { list. inner HTML += `
```

```

<div class="course-card">
  <h4>${course.name}</h4>
  <p>Status:
    <span class="${course.completed ? "completed" : ""}">
      ${course.completed ? "Completed" : "Pending"}
    </span>
  </p>
  <button onclick="toggleCourse(${course.id})">
    Mark as ${course.completed ? "completed" : "pending"}
  </button>
</div>
  `;
});

```

```

    updateProgress();
  }

```

```

// ===== ADD ACTIVITY (COURSE PAGE) =====

```

```

function addActivity() {
  const input = document.getElementById("activityInput");
  const name = input.value.trim();

  if (name === "") {
    alert("Please enter activity name");
    return;
  }

```

```

  courses.push({ id
    : Date.now(),

```

```

    name: name,
    completed: false
  });

```

```

    input.value = "";
    localStorage.setItem("courses", JSON.stringify(courses));
    displayCourses();
  }

```

// ===== ADD ACTIVITY FROM HOME =====

```

function addActivityFromHome() {
  const input = document.getElementById("homeActivityInput");
  const name = input.value.trim();

  if (name === "") {
    alert("Please enter activity name");
    return;
  }

```

```

    courses.push({ id
      : Date.now(),
      name: name,
      completed: false
    });

```

```

    input.value = "";
    localStorage.setItem("courses", JSON.stringify(courses));
    loadHome();
  }

```


// ===== TOGGLE =====

function toggleCourse(id)

{ courses = courses.map(c =>

c.id === id ? { ...c, completed: !c.completed } : c

);

localStorage.setItem("courses", JSON.stringify(courses));

displayCourses();

}

// ===== PROGRESS =====

function updateProgress() {

const completed = courses.filter(c => c.completed).length;

const percent = courses.length === 0

? 0

*: Math.round((completed / courses.length) * 100);*

document.getElementById("progressText").innerText =

percent + "% Completed";

document.getElementById("progressFill").style.width =

percent + "%";

}

// ===== RESET =====

function resetCourses() {

courses.forEach(c => c.completed = false);

localStorage.setItem("courses", JSON.stringify(courses));

```
displayCourses();
}

// ===== INITIAL LOAD =====

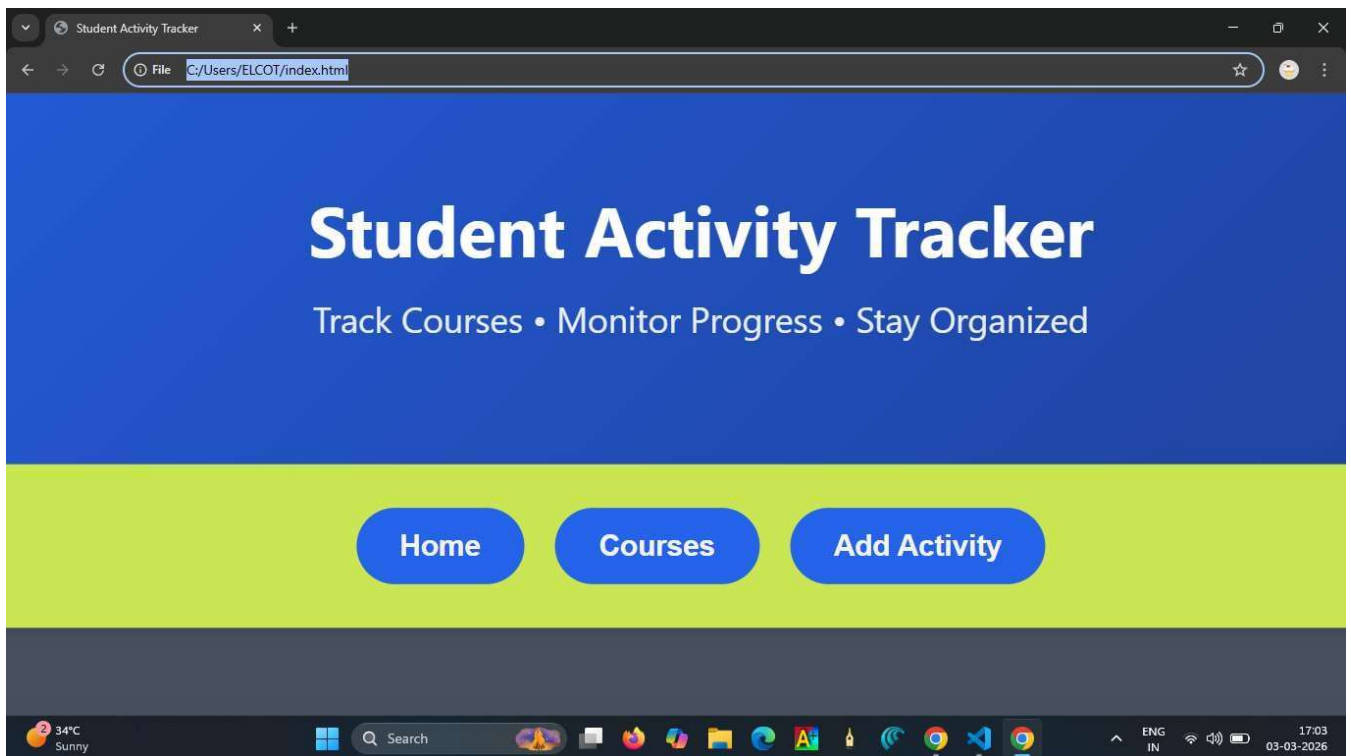
loadHome();
showHome();
```

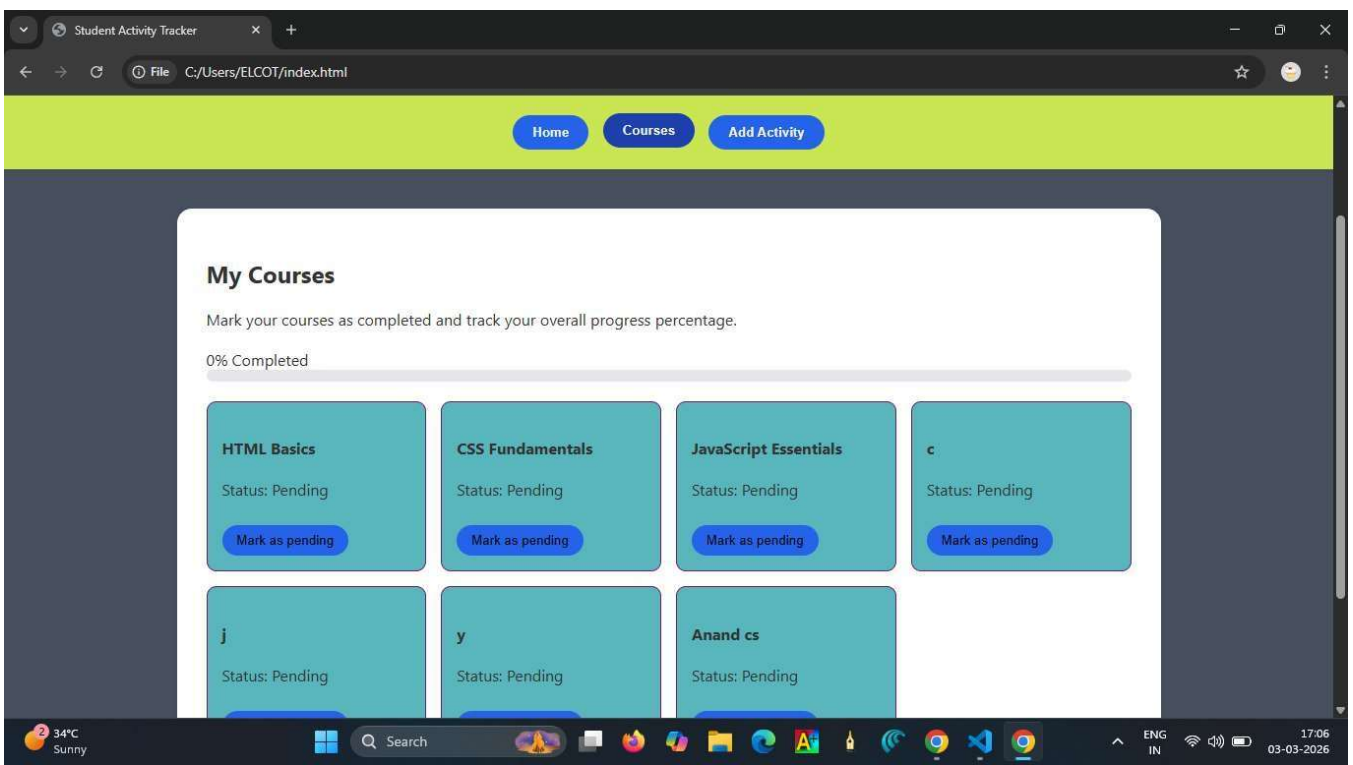
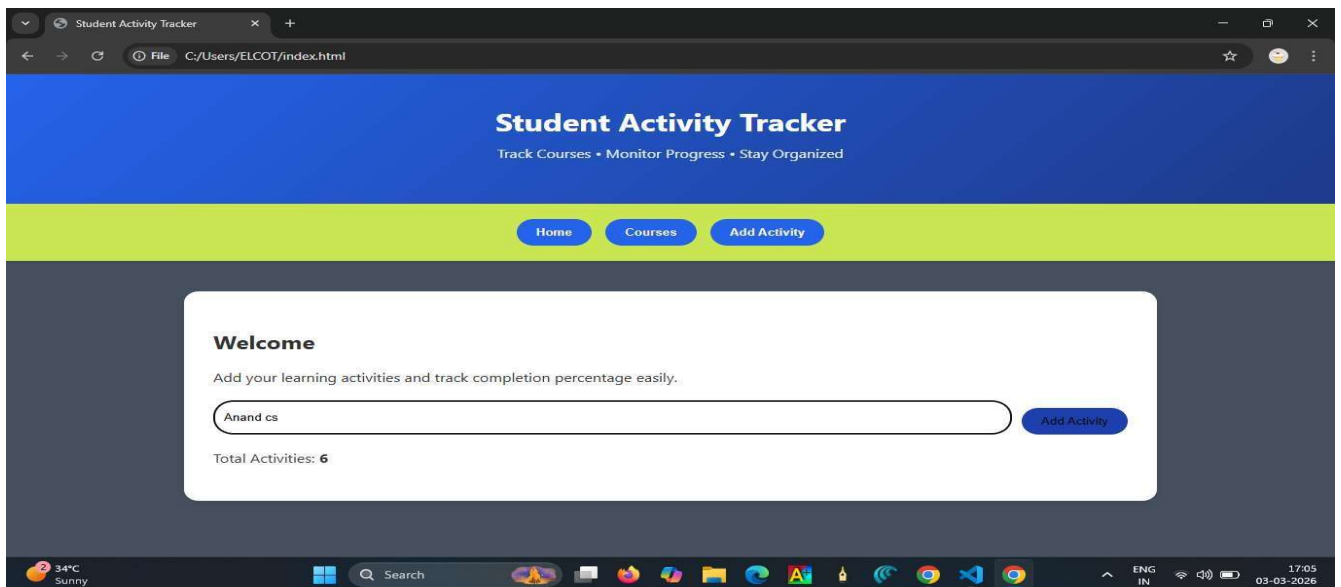
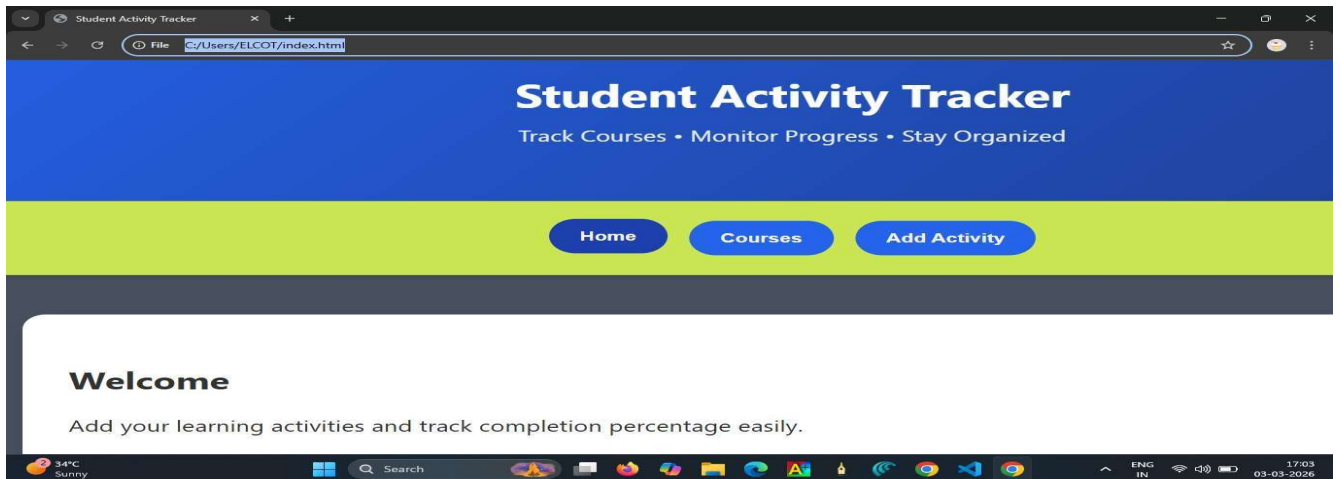
10. Expected Outcome

The final result is a simple and user-friendly web application that:

- *Allows students to track their activities*
- *Helps monitor progress visually*
- *Provides practical learning experience in front-end development*

11. output





Student Activity Tracker

File C:/Users/ELCOT/index.html

My Courses

Mark your courses as completed and track your overall progress percentage.

57% Completed

HTML Basics Status: Completed Mark as completed	CSS Fundamentals Status: Pending Mark as pending	JavaScript Essentials Status: Pending Mark as pending	c Status: Completed Mark as completed
j Status: Completed Mark as completed	y Status: Pending Mark as pending	Anand cs Status: Completed Mark as completed	

Reset All Courses

34°C Sunny
Search
17:07 03-03-2026

My Courses

Mark your courses as completed and track your overall progress percentage.

0% Completed

HTML Basics Status: Pending Mark as pending	CSS Fundamentals Status: Pending Mark as pending	JavaScript Essentials Status: Pending Mark as pending	c Status: Pending Mark as pending
j Status: Pending Mark as pending	y Status: Pending Mark as pending	Anand cs Status: Pending Mark as pending	

Reset All Courses

12. Learning Outcome

Through this project, the following skills are gained:

- *UI Design principles*
- *HTML structure creation*
- *CSS styling and responsive layout*
- *JavaScript DOM manipulation*
- *Local storage usage*
- *Basic application planning and usability design*

13. Conclusion

- *The Student Activity Tracker successfully fulfills the objective of creating a lightweight, front-end-only web application. It provides a simple yet effective way for students to track their learning activities without backend complexity.*
- *The project enhances front-end development skills while delivering a practical and useful solution.*